

Adaptive Charging Threshold and Detour Scheduling for Wireless Rechargeable Sensor Networks via Deep Reinforcement Learning

Liang-Ching Tao¹, Wei-Cheng Tai¹ and Pi-Chung Wang^{1,*}

¹ Department of Computer Science and Engineering, National Chung Hsing University, Taichung 402, Taiwan; liangkingtao@gmail.com; allentai84@gmail.com

* Correspondence: pcwang.tw@gmail.com

Problem. On-demand charging in wireless rechargeable sensor networks lets a node request service once its energy drops below a threshold θ . Nearly all schemes fix one θ network-wide, ignoring how unevenly relay load falls across clusters: set it high and the charger queue saturates, set it low and nodes die before service arrives [1]. Dispatch is equally rigid: chargers run strict base-target-base trips, blind to an urgent cluster beside the route [2].

Approach. We add two cooperating deep-RL agents to a conventional clustering-and-routing stack. A **threshold agent** (a DQN with a GRU) resets θ per cluster every 60 s from recent request history; a **route agent** (MLP-DQN) decides at departure whether to detour to a critical cluster en route. A two-stage curriculum trains the threshold agent first, then freezes it while the route agent learns.

Results. Over 12 matched energy seeds at each of five network sizes, the framework beats the fixed-threshold scheme it builds on at every size, by **2.6 to 14.2** percentage points, and every other baseline except MCWP STO-CS at the lightest load (3.1 points higher). The spread matters more than the means: at 200 sensors, 32 seeds put the fixed threshold in a bimodal regime—**8 collapses** (25%), the worst leaving 9% alive—while the adaptive threshold spans only 89.0–94.5% ($\sigma = 1.6$) there and never drops below **86.3%** at any size. Adapting θ removes a failure mode rather than shifting an average.

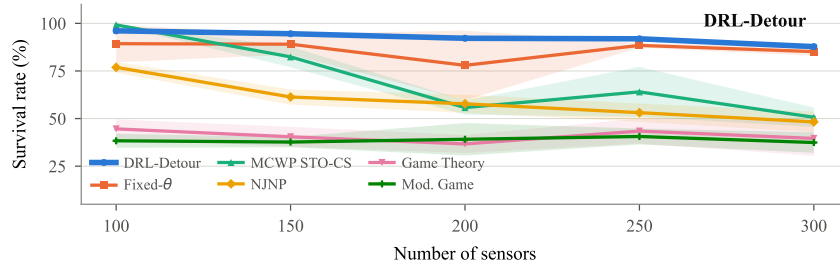


Fig. 1. Survival rate versus sensor count; bands are ± 1 standard deviation over 12–15 seeds per point (32 for the fixed threshold at 200 sensors). The wide fixed-threshold band marks real instability, not sampling noise.

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[1] L. He *et al.*, “Evaluating the on-demand mobile charging in WSNs,” *IEEE Trans. Mobile Comput.*, 14(9), 1861–1875, 2015.

[2] K. Wang, “Dynamic on-demand collaborative charging scheduling based on game theory,” *IEEE J. Electron Devices Soc.*, 12, 824–830, 2023.